WHY WATER TREATMENT IS REQUIRED ?

ACTIVITIES OF CHEMISTRY

- CHEMISTRY
  - COAL & OIL
  - MAIN PLANT CHEMISTRY
  - WATER TREATMENT PLANT
  - PURCHASE,STORE ENVIRONMENT SPECIAL LAB R&M

USE OF WATER

- **AS COOLING WATER FOR CONDENSER (RAW WATER)**
- **AS COOLING WATER UNIT AUXILIARIES (CLF WATER)**
- **AS COOLING WATER OF PRIMARY WATER(CLF WATER)**
- **DM WATER AS MAKE UP WATER**
- **DRINKING WATER (FILTERED CLARIFIED WATER)**
- **LP/HP WATER**

OBJECTIVES OF WATER CHEMISTRY CONTROL

Maintain the plant in a condition where it can operate reliably and smoothly.

- Maintain the integrity of feed, boiler and steam systems
- Maintain the effectiveness of heat transfer surface
- Maintain the high level of steam purity

WATER IMPURITIES

SUSPENDED SOLIDS

- MUDS
- SILTS
- SAND
- ORGANICS

DISSOLVED SOLIDS

- CALCIUM SALTS
- MAGNESIUM SALT
- SODIUM SALTS
- IRON, MANGANESE, FLUORIDE SALTS
- GASES

METHOD OF TREATMENT

RAW WATER → PRETREATMENT

TO REMOVE SUSPENDED SOLIDS

CLARIFIED WATER

FILTRATION

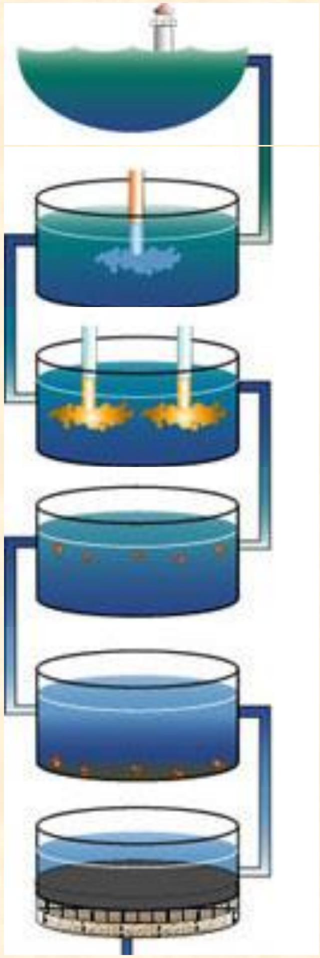
TO REMOVE DISSOLVED SOLIDS → DEMINERALISATION

DM WATER

CHEMICALS USED IN TREATMENT

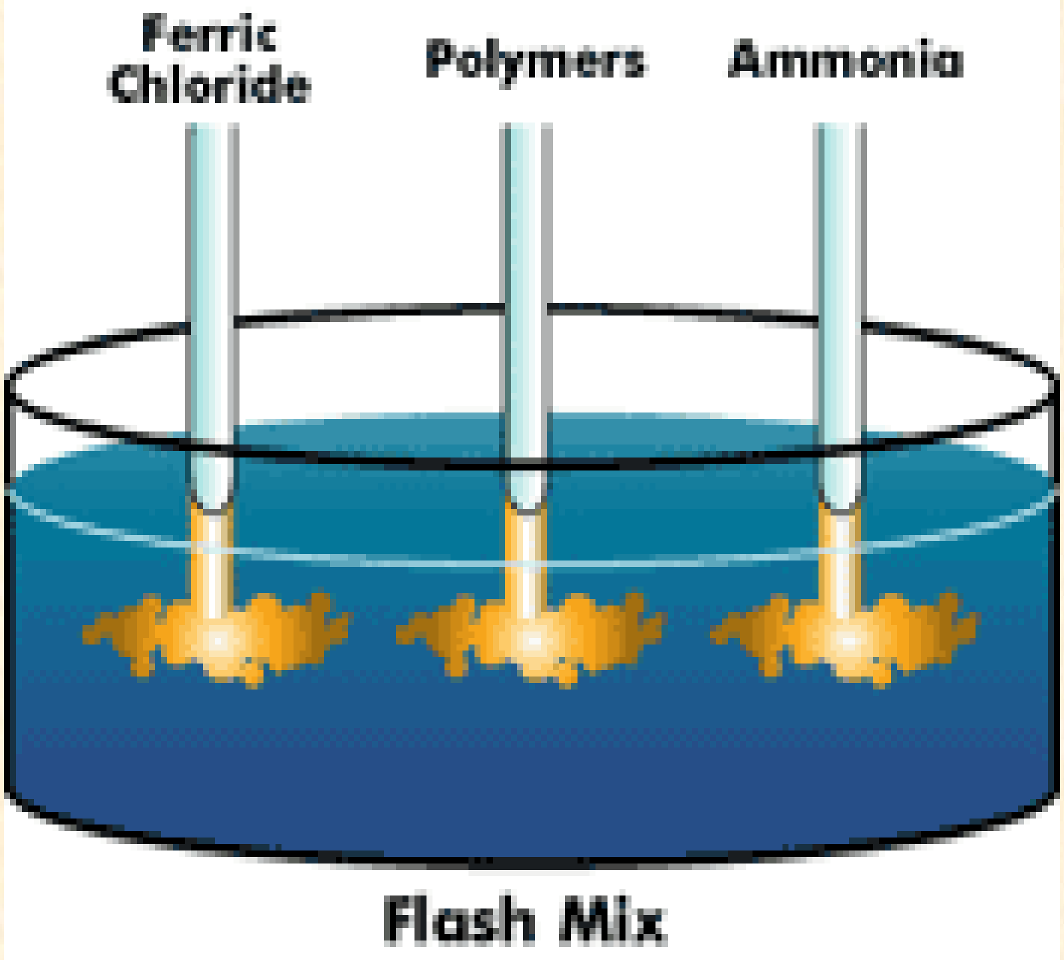
SL.NO	CHEMICALS	USE
1	ACF	TASTE,ODOUR & DECHLORINATION
2	ALUM, $FeCl_{3,}$ PE	COAGULANT
3	BENTONITE	FLOC WEIGHING AGENT
4	LIME	pH ADJUSTMENT, SOFTENING
5	Cl_2, ClO_2, O_3, Ca(OCl)Cl	DISINFECTION, TO CONTROL BACTERIA, VIRUS & ORG. MAT
6	HYDRAZINE	OXYGEN SCAVENGER, MAGENTITE LAYER
7	AMMONIA	pH ADJUSTMENT
8	TSP	TO TREAT BOILER WATER
9	HCl ACID	REGENERANT OF CATION RESIN
10	NaOH	REGENERANT OF ANION RESIN

PRETREATMENT PROCESS



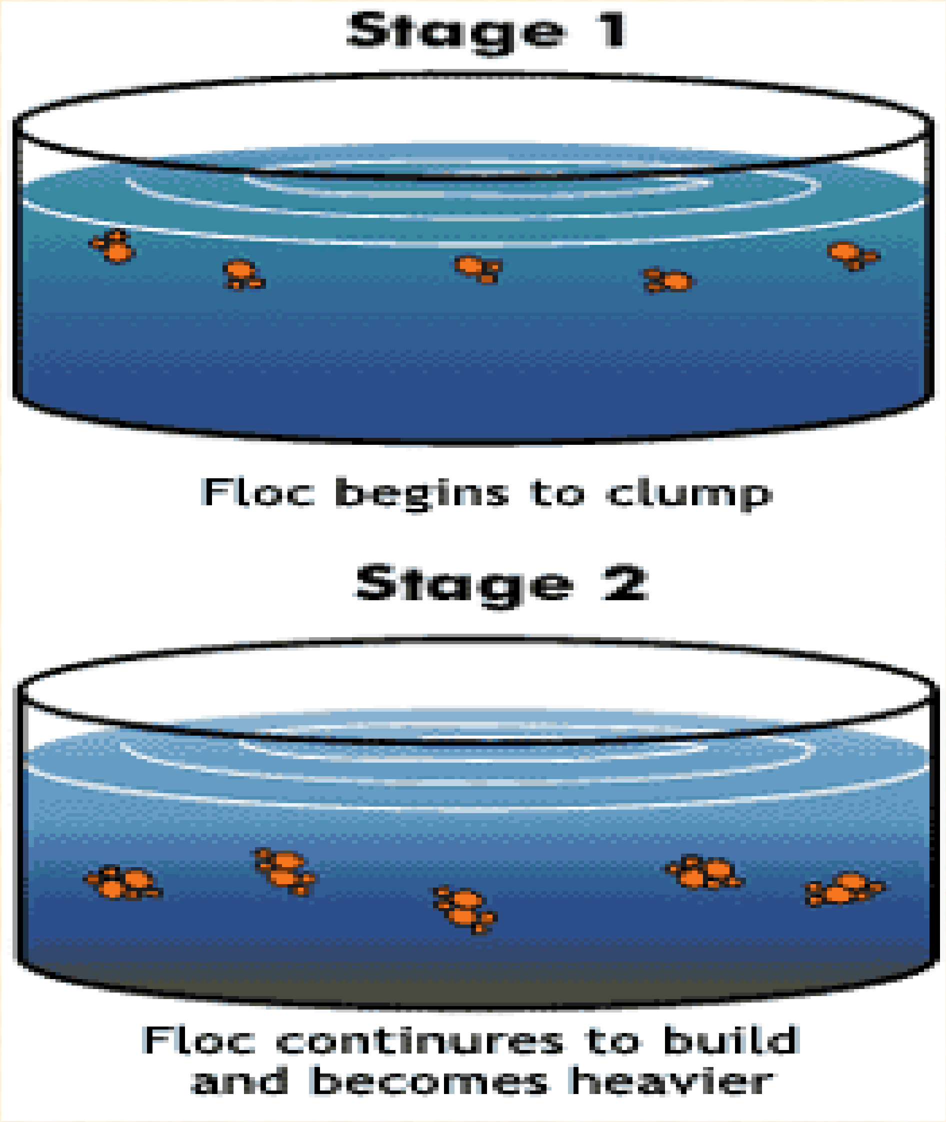
Sweetwater
Reservoir
↓
Disinfection
↓
Coagulation
↓
Flocculation
↓
Sedimentation
↓
Filtration → **CLEAR WATER**

COAGULATION



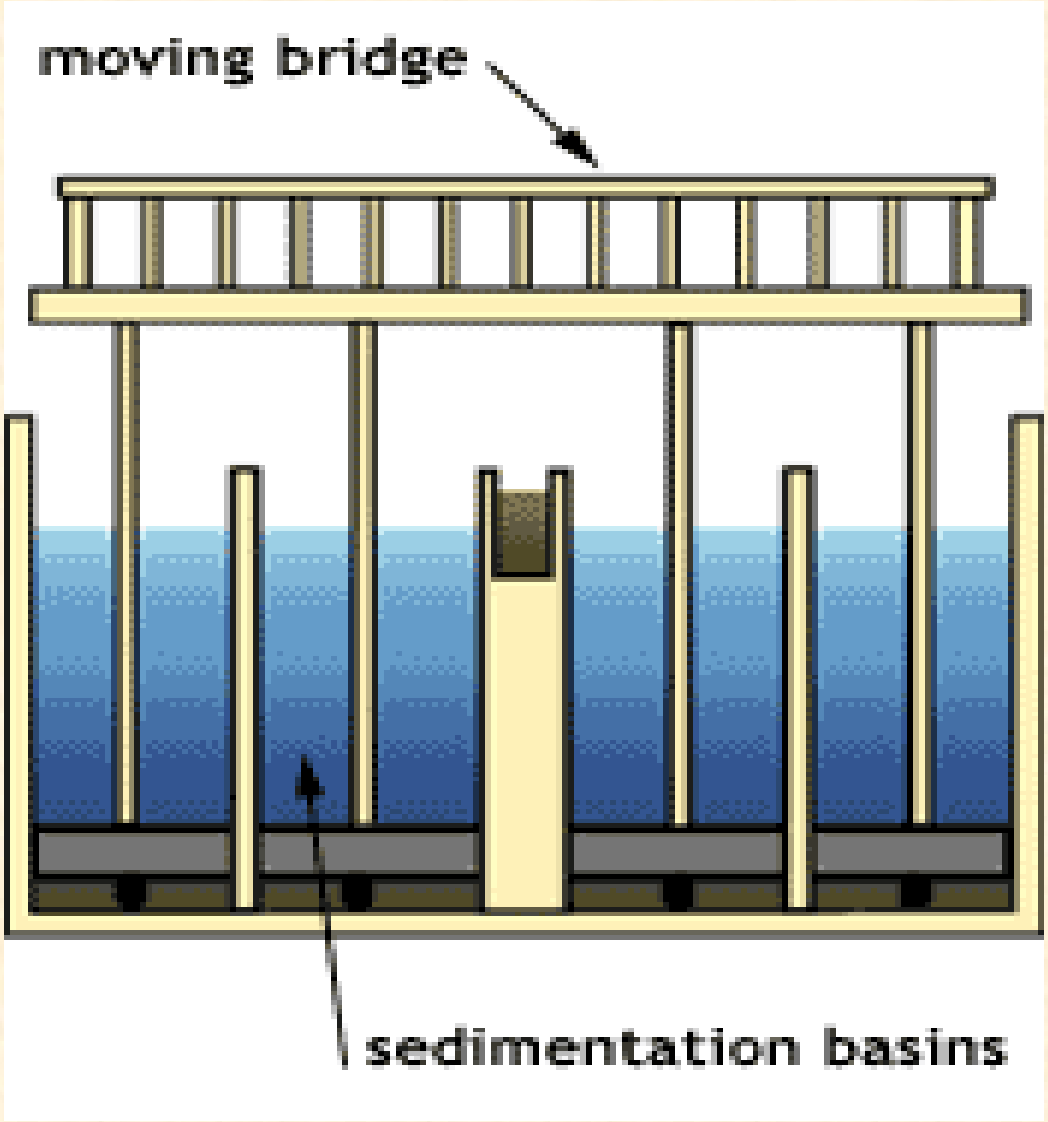
Lime, Aluminium sulphate are quickly added and mixed into the water to begin the process of removing dirt and particles.

FLOCCULATION



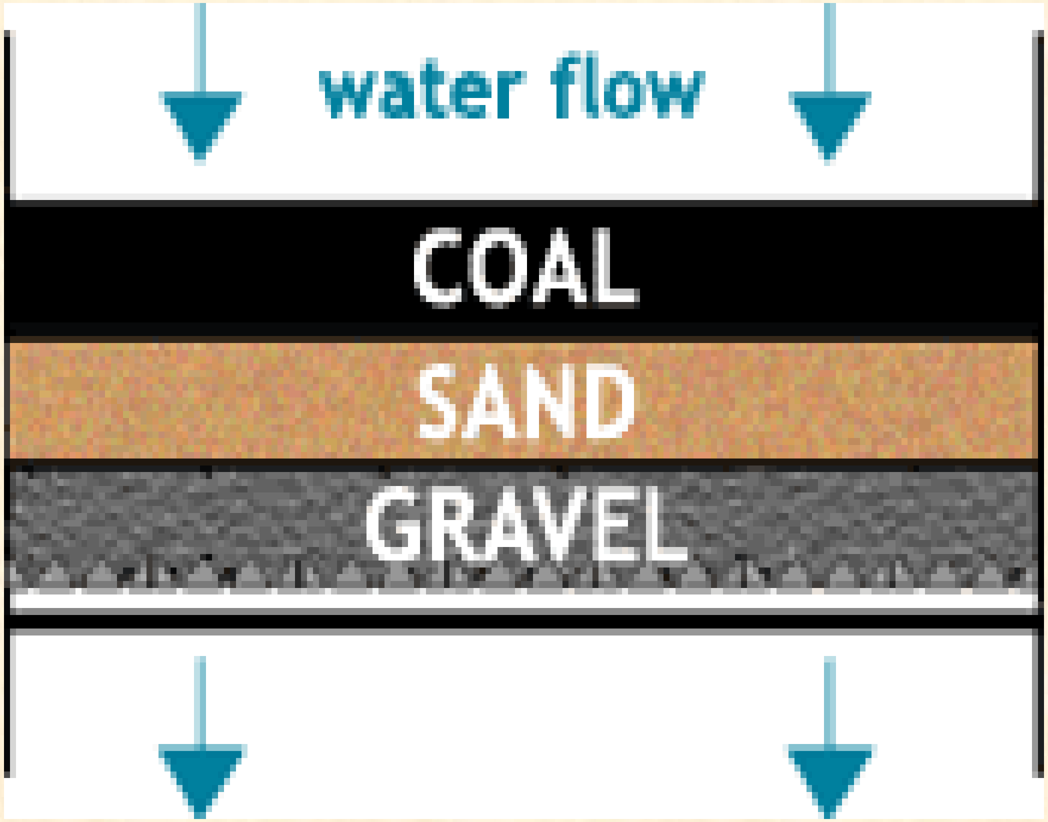
Gentle mixing of the Aluminium Sulphate and Lime in the water causes dirt particles to stick together and become heavy, forming rust-colored "floc"

SEDIMENTATION



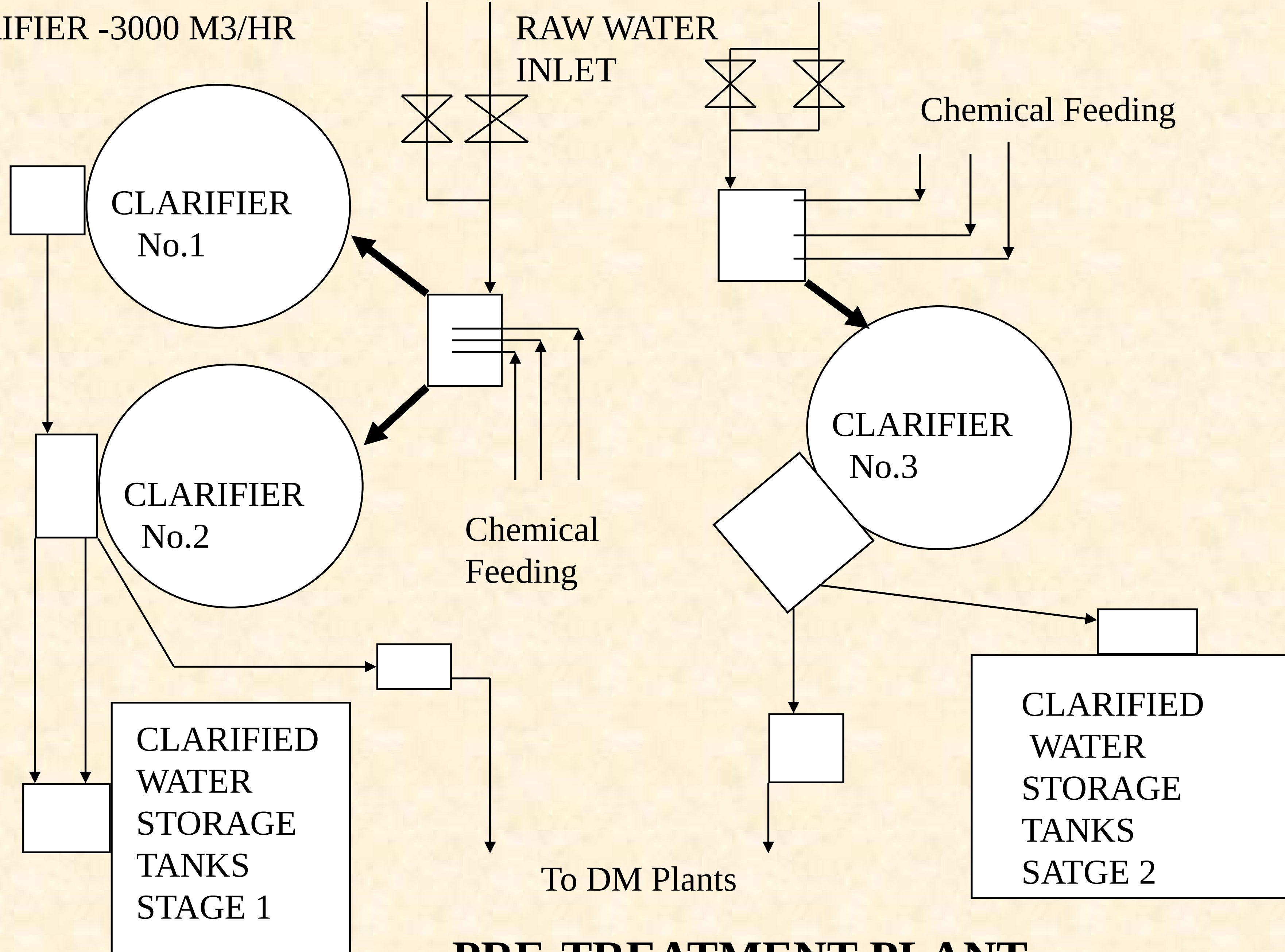
The heavy, sticky floc clumps settle to the bottom of the basins, where they are vacuumed up by the moving bridge. The remaining water flows over a raised baffle, then moves to the filters.

FILTRATION



The water passes down through filters (layers of anthracite coal, sand and gravel) which trap and remove the remaining particles

CLARIFIER -3000 M3/HR

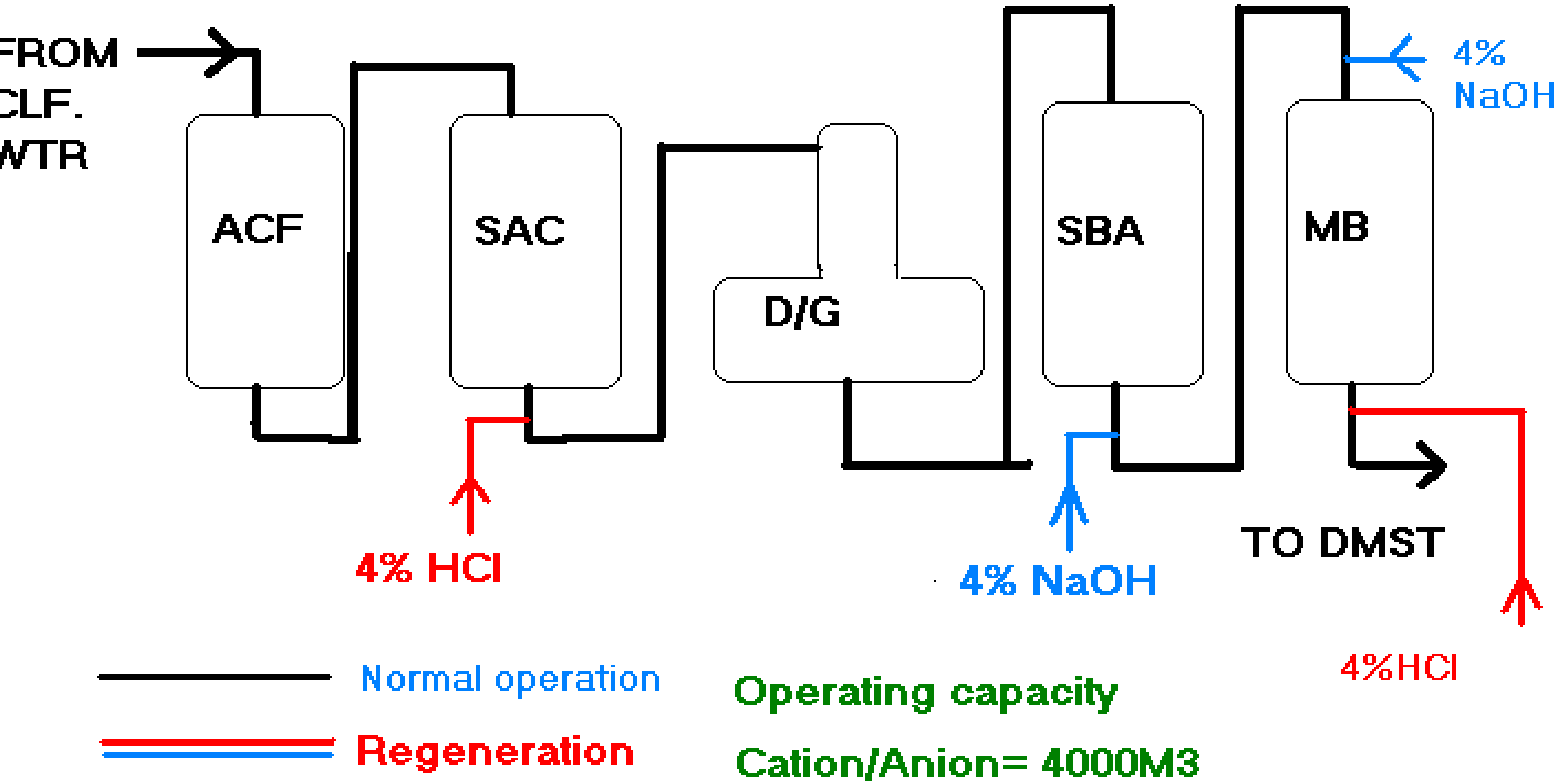


PRE-TREATMENT PLANT

STANDARD FOR CHEMICAL CONTROL

SL NO	ITEM DESCRIPTION	pH	TURBI DITY/ K	SILICA
1	DRINKING WATER	6.5-7.5	5-10NTU	
2	CLARIFIED WATER	6.5-7.5	15-20 NTU	
3	DEMINERALISED WATER	6.8-7.0	K= 0.2-0.1	10-20 PPB

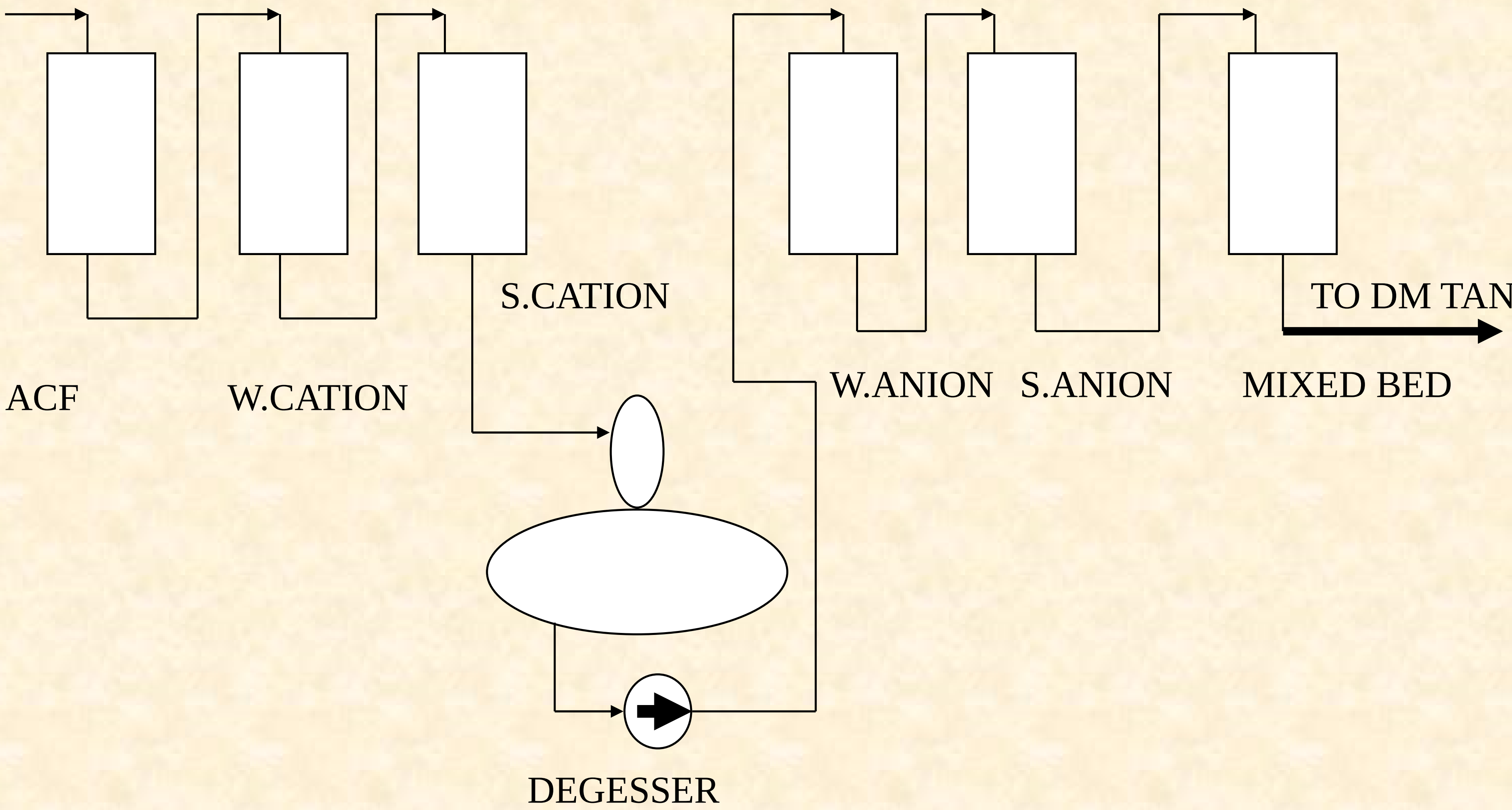
DEMINERALIZATION



Operating capacity

Cation/Anion= 4000M3

MB = 15000M3

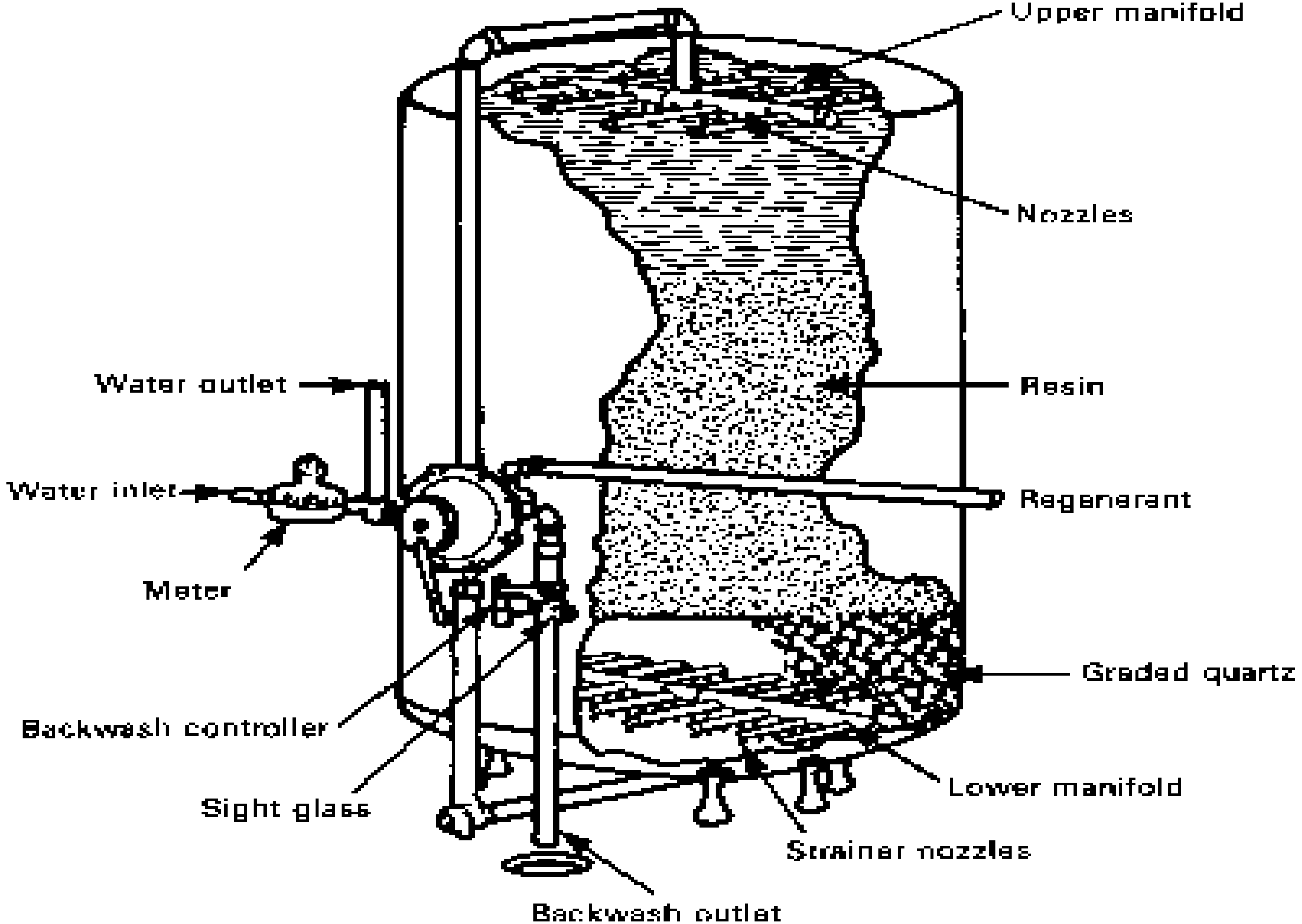


D.M.STREAMS: D.M.PLANT

NOS. OF STREAM: 2+1

CAPACITY : 100 M3/HR Each

VIEW OF AN ION EXCHNGE VESSEL

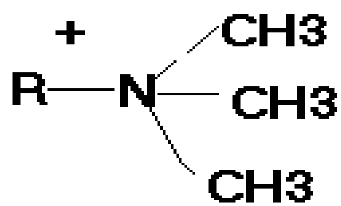


SOURCE: Kunin, R. "Ion Exchange for the Metal Products Finishers." (3 pts.), *Products Finishing*. Apr.-May-June 1969.

DEMINERALISATION

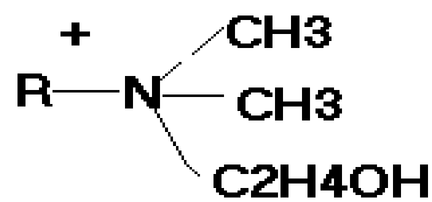
STRONG BASE ANION RESIN

TYPE I



FOR SILICA

TYPE II

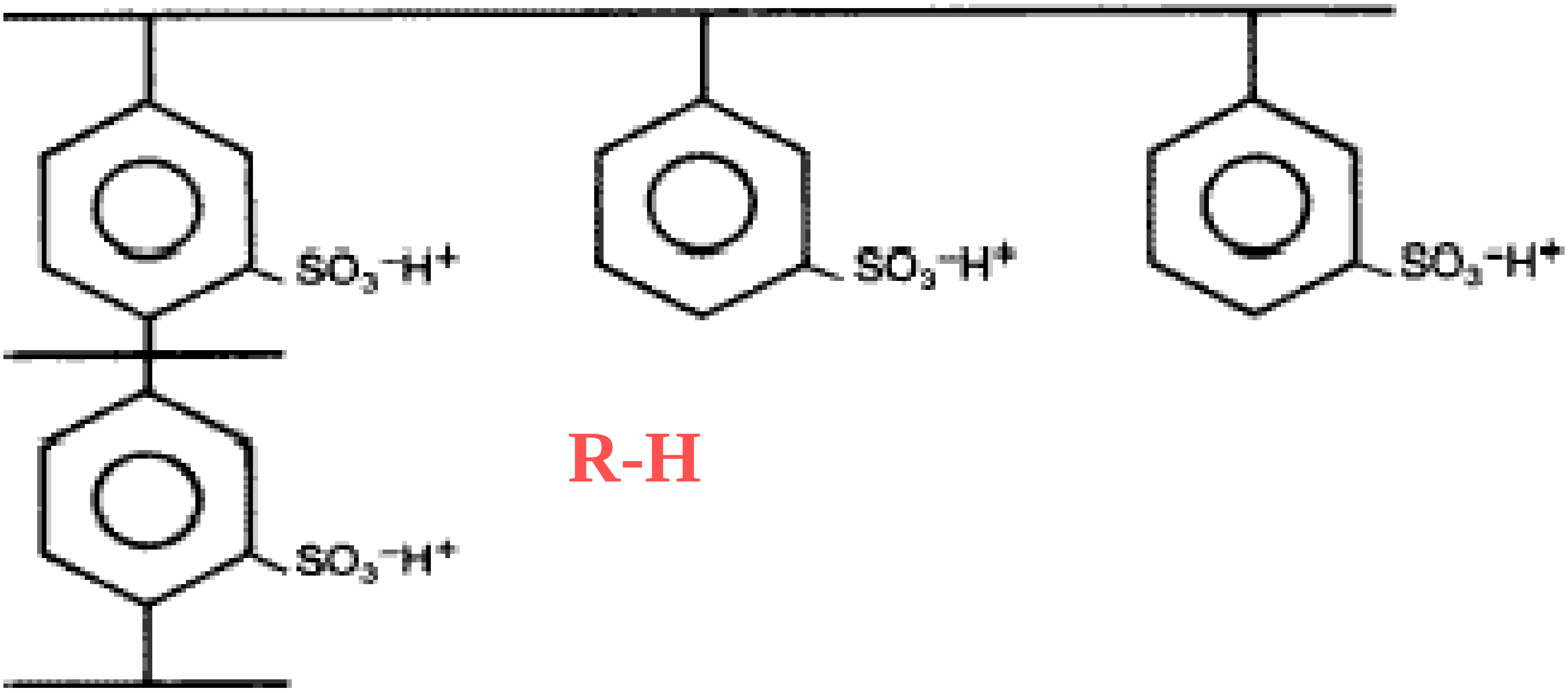


FOR BETTER REGN

R = Polymer (crosslinked) chain

Functional groups = quaternary Ammonium groups

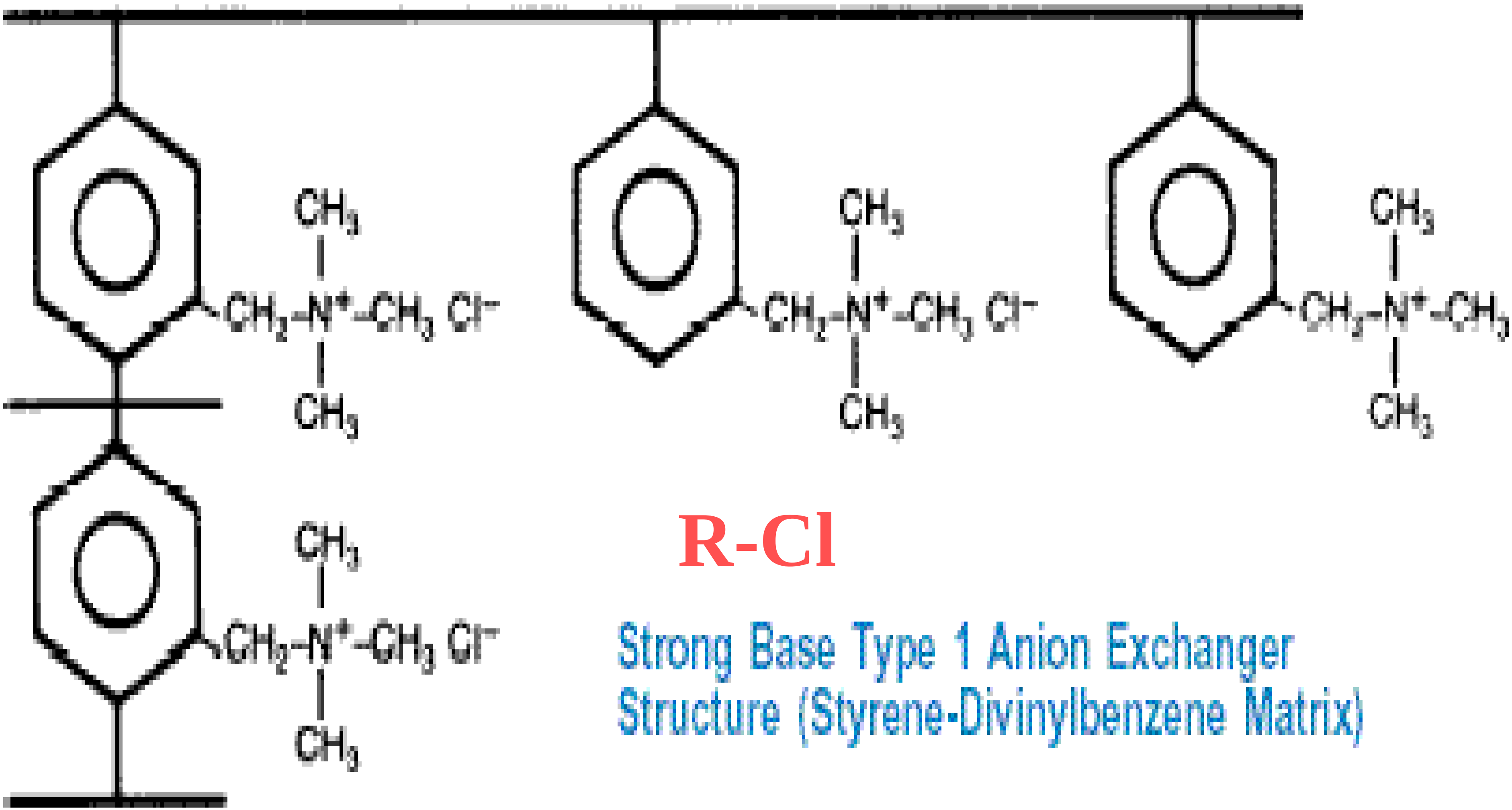
STRUCTURE OF CATION RESIN



R-H

Strong Acidic Cation Exchanger Structure
(Styrene-Divinylbenzene Matrix)

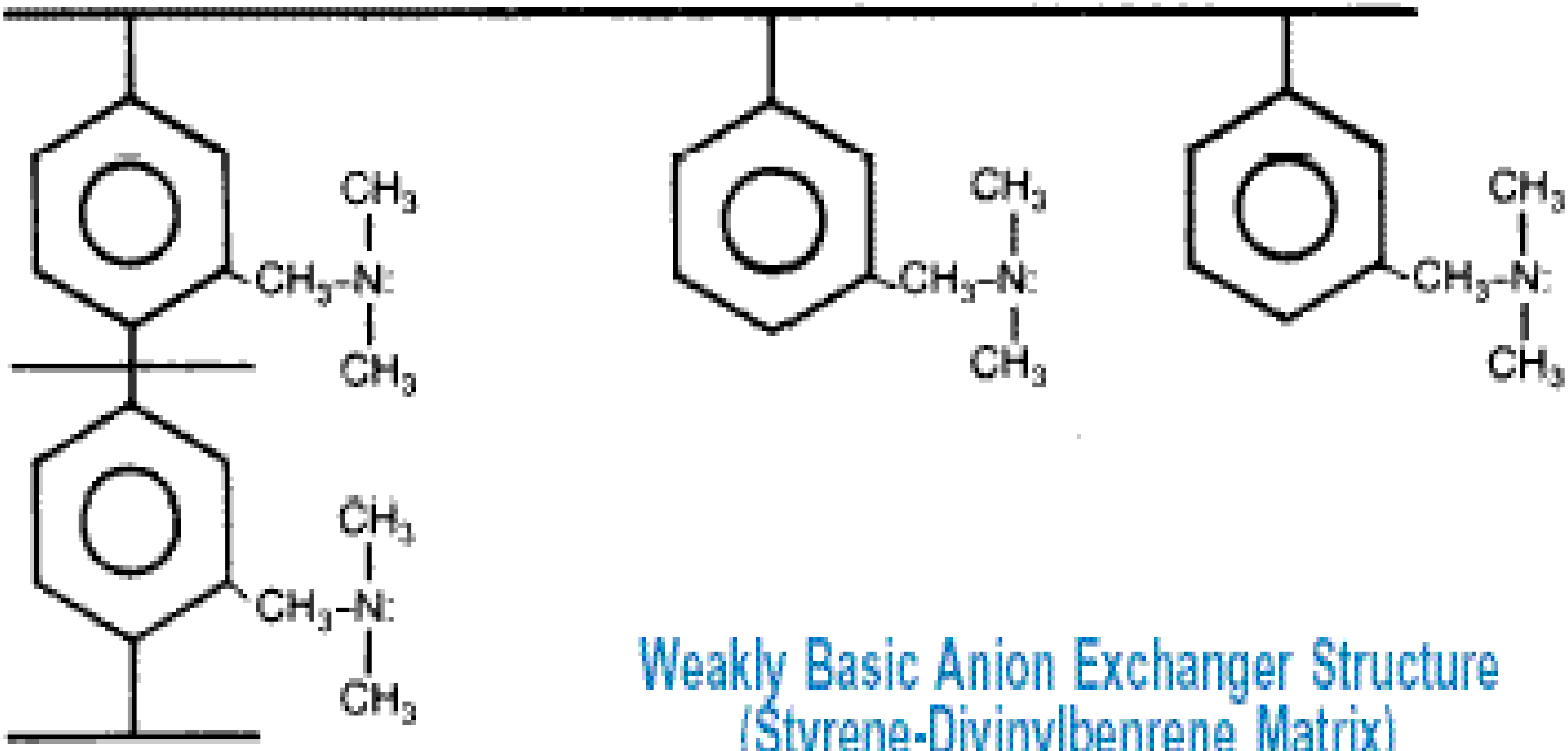
STRUCTURE OF ANION RESIN



R-Cl

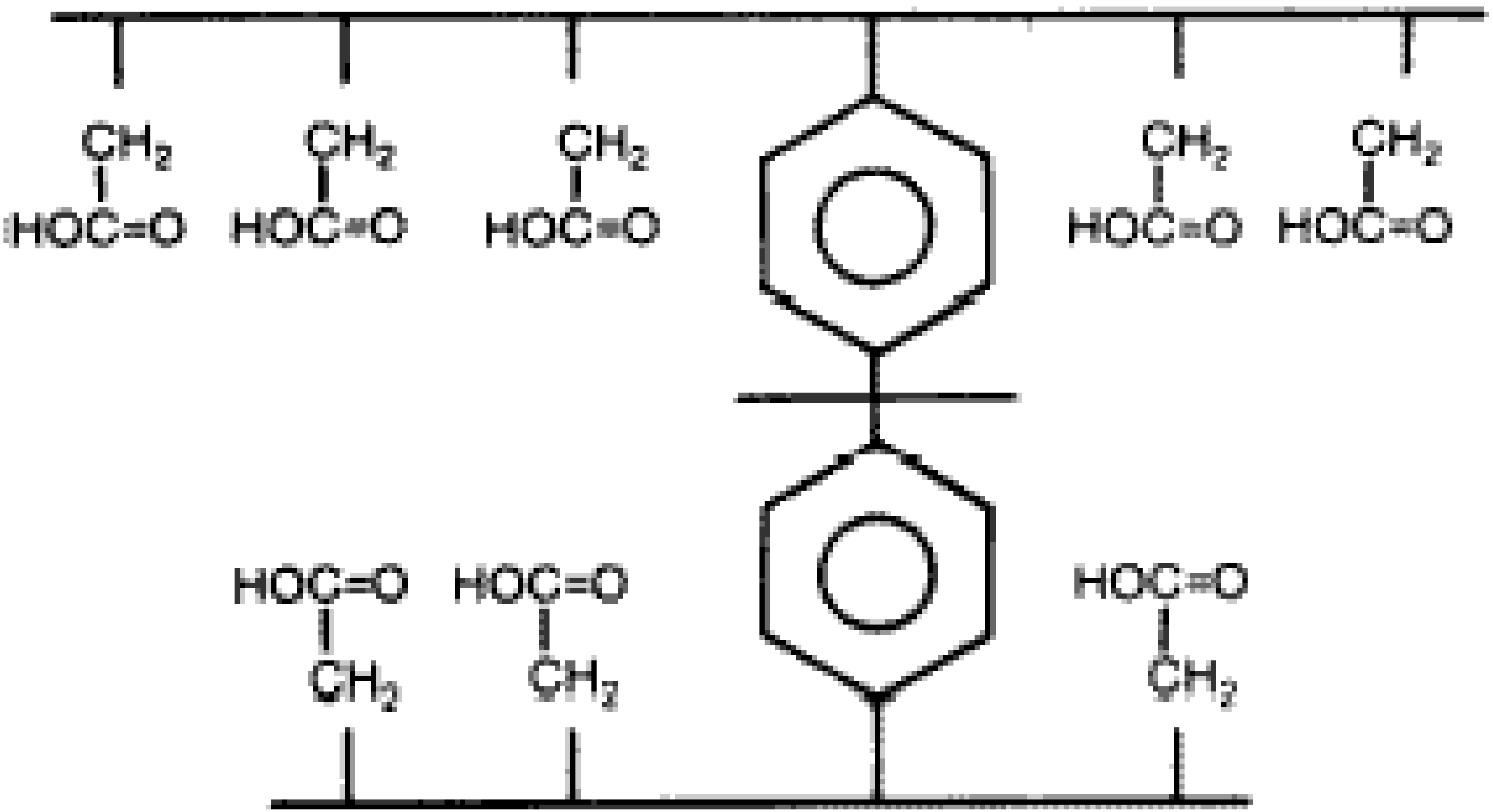
Strong Base Type 1 Anion Exchanger Structure (Styrene-Divinylbenzene Matrix)

STRUCTURE OF WBA RESIN



Weakly Basic Anion Exchanger Structure
(Styrene-Divinylbenrene Matrix)

STRUCTURE OF WAC RESIN



Weakly Acidic Cation Exchanger Structure
(Acrylic Divinylbenzene Matrix)

ION EXCHANGE REACTIONS

DEMINERALISATION CATION EXCHANGE

$$H_2R + \left.\begin{array}{l} K_2 \\ Na_2 \\ Ca \\ Mg \\ (NH_4)_2 \\ Fe \end{array}\right| \begin{array}{l} SO_4 \\ Cl_2 \\ F_2 \\ (NO_3)_2 \end{array} \rightleftharpoons \left.\begin{array}{l} K_2 \\ Na_2 \\ Ca \\ Mg \\ (NH_4)_2 \\ Fe \end{array}\right| R + \begin{array}{l} H_2SO_4 \\ 2\,HCl \\ 2HF \\ 2HNO_3 \end{array}$$

Neutral salts

Strong Acids

Resin in H+ form

Resin in cation form

ION EXCHANGE REACTIONS

ANION EXCHANGE

$$R(OH)_2 + \begin{matrix} H_2SO_4 \\ 2\,HCl \\ 2HF \\ 2HNO_3 \end{matrix} \rightleftharpoons R\begin{matrix} SO_4 \\ Cl_2 \\ F_2 \\ (NO_3)_2 \end{matrix} + 2H_2O$$

Strong Acids

Water

Resin in OH form

Resin in anion form

REGENERTION

CATION RESIN REGENERATION REACTION

$$R \left| \begin{array}{l} K_2 \\ Na_2 \\ Ca \\ Mg \\ (NH_4)_2 \\ Fe \end{array} \right. + HCl \rightleftharpoons R\text{-}H + \begin{array}{l} 2KCl \\ 2NaCl \\ CaCl_2 \\ MgCl_2 \\ (NH_4)_2Cl \\ FeCl_2 \end{array}$$

4% HCl

Neutral salts

Resin in cation form

Resin in H+ form

REGENERATION

REGENERATION OF ANION RESIN

$$R\left|\begin{array}{l} SO_4 \\ \\ Cl_2 \\ F_2 \\ (NO_3)_2 \end{array}\right. + NaOH \rightleftharpoons R(OH)_2 + \begin{array}{l} Na_2SO_4 \\ 2NaCl \\ 2NaF \\ 2NaNO_3 \end{array}$$

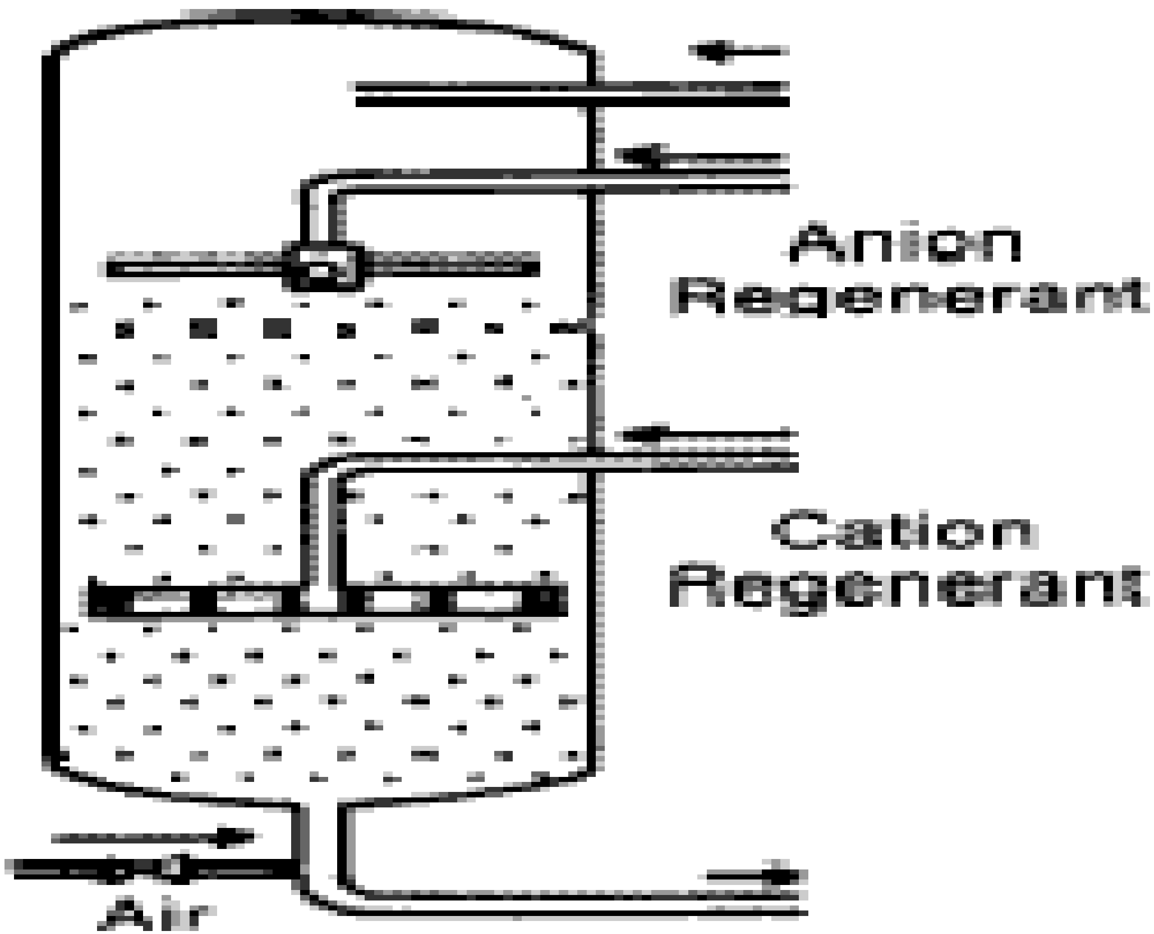
4% NaOH

Neutral salts

Resin in Anion form

Resin in OH form

REGENERATION OF MIXED BED



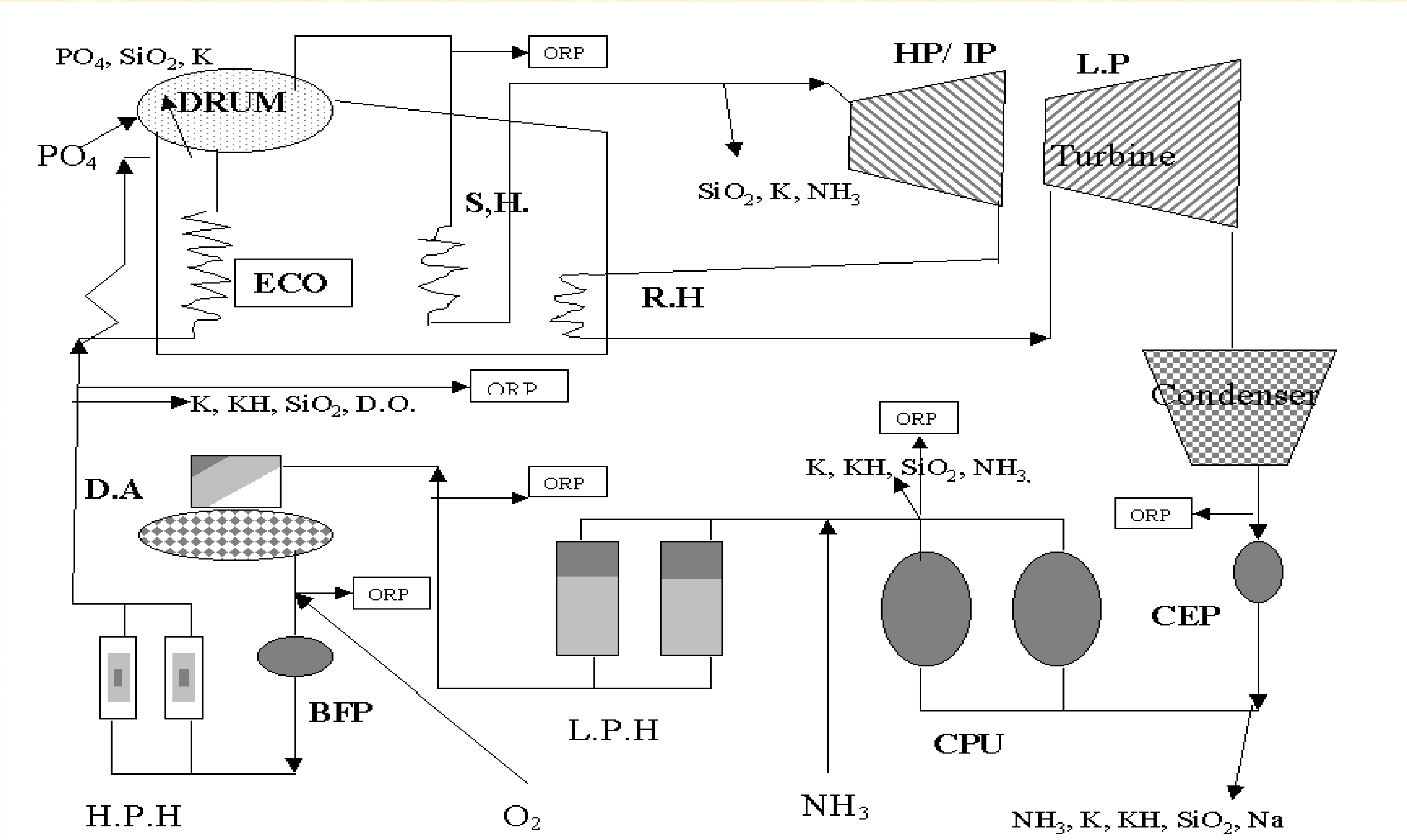
Mixed Bed Ion Exchange Unit

BOILER WATER TREATMENT

RECOMMENDED LIMIT OF SILICA IN BOILER WATER AT PH 9.5 AT DIFF OPERATING PRESSURE TO LIMIT SILICA IN STEAM

DRUM PR	BLR SILICA	DRUM PR	BLR SILICA
60	8.2	150	0.4
80	3.8	155	0.37
100	1.9	160	0.30
120	0.9	165	0.27 (200MW)
130	0.7	170	0.25
140	0.52	175	0.22
145	0.48	180	0.18(500MW)

CYCLE CHEMISTRY CONTROL AND INSTRUMENTATION



THANK YOU